Runtime Security for Agentic Systems

A Practical Two-Plane Approach for OpenClaw-Class Agents

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Abstract

Tool-using AI agents expose a runtime security gap: pre-install scanning and configuration hardening are necessary, but they cannot fully characterize behavior that emerges during execution (e.g., indirect prompt injection and tool poisoning). We present a two-plane security model that correlates reasoning-plane observations (independently read session transcripts) with system-plane telemetry to detect intent-behavior divergence.

The contribution is architectural: process-attributed telemetry, task-scoped intent declarations, and a confidence-stratified signal hierarchy form a portable control loop across MCP-compatible stacks. In the reference implementation, EDAMAME accepts agent-tagged behavioral-model slices from multiple reasoning-plane producers, including OpenClaw and Cursor, merges them into one observer-owned correlation window, and preserves contributor attribution. To demonstrate feasibility and relevance, the live benchmark in this draft uses an OpenClaw-derived scenario corpus and the explicit-slice producer path (`upsert_behavioral_model`) to exercise EDAMAME's downstream correlation logic, while the same EDAMAME control loop also supports raw-session ingest (`upsert_behavioral_model_from_raw_sessions`) for Cursor on a developer workstation.

On the current canonical BadAgentUse artifact lineage (50 live scenarios across six benchmark categories, plus two integration-validated categories; 273 valid runs across 12 seeds), precision is 0.862 (CI95: [0.815, 0.899]) and recall is 0.987 (CI95: [0.962, 0.996]). Median detection latency is 45.8 s (p95: 53.6 s): fast enough to support many operator responses, but still a meaningful pre-alert window and therefore the primary limitation of a reactive design. These aggregate metrics are cross-checked by layered model-to-verdict validation of the core engine and a focused live validation ladder spanning clean, unexpected-egress, `/tmp` lineage, and designed-miss behavior. We explicitly mark designed blind spots: living-off-the-land behavior under allowed process/destination pairs and timing-window evasion before telemetry arrives. This frames the model as a practical correlation architecture with an explicit latency trade-off versus interception-first systems such as AgentSentinel, while still providing a portable and trace-auditable runtime detection path.

Keywords

agent security, runtime monitoring, prompt injection, tool poisoning, Model Context Protocol (MCP), intent-behavior correlation, telemetry, reproducibility

1. Introduction

Agent systems now perform high-impact operations through tools and MCP integrations: network access, file mutation, credential-adjacent actions, and persistent memory updates. Risk therefore depends on what the agent actually does at runtime, not only on what is statically configured before runtime.

A recent publicly discussed incident involving Meta security leadership and an OpenClaw deployment that reportedly executed unintended mailbox-deletion actions illustrates the practical problem: loss of control can happen during normal runtime, even in technically sophisticated environments. Our model targets that runtime-loss-of-control phase by providing independent host-side observation and divergence alerts. It does not replace authorization design or pre-action approval gates; if broad destructive permissions are already granted, prevention still depends on those controls.

We characterize the problem in terms of three **control strata** to avoid terminology collision with our model's two planes:

1. **S1: Supply-chain controls** (e.g., signed artifacts, provenance/SBOM checks).
2. **S2: Configuration controls** (e.g., localhost MCP bind + PSK auth, role and tool scoping, sandboxing, policy hardening).
3. **S3: Runtime behavior controls** (e.g., a declared read-only task followed by undeclared egress, detected via intent-behavior consistency checks).

The residual runtime-only failures that remain after S1/S2 controls are what we call the **S3 gap**.

Together, S1/S2 can stop many failures. For example, an internet-exposed OpenClaw gateway (SecurityScorecard STRIKE, 2026) is often prevented by correct bind/auth configuration, which is fundamentally a configuration-control problem. However, S1/S2 do not fully address runtime divergence: in CVE-2026-25253, a browser token exfiltration path can execute during normal user interaction even when baseline setup appears acceptable. Similar runtime-only gaps arise in coding agents operating over production repositories, where unauthorized access to git-managed secrets, SSH keys, or CI tokens can produce outsized impact.

This paper addresses the S3 gap with a two-plane model: compare what the reasoning plane declares with what the system plane observes on the host (defined in Section 3). The approach highlights a structural opportunity for security tooling vendors (here, EDAMAME Posture) and AI platform providers (here,

OpenClaw and Cursor) to collaborate through MCP: the security tool provides system-plane ground truth that no AI framework includes natively, while the AI platform provides the runtime observation surface (session transcripts, memory, tool invocation history) that no external monitor can access alone. Neither plane is sufficient in isolation; the two-plane correlation is the security primitive. By design, this first-generation rule set has a blind spot for living-off-the-land behavior when both process and destination are already authorized; we surface that limitation explicitly in Sections 7 and 9.

1.1 Contributions (Novelty)

We separate contributions into a model part and an implementation/evaluation part.

Model contributions (Part I):

- **Two-plane runtime security model.** A portable control loop for correlating observed agent behavior (reasoning plane, via session transcript observation) with independent host telemetry (system plane).
- **Model-level requirements.** Process-attributed telemetry and a signal hierarchy are presented as requirements of the model, not as product-specific implementation details.
- **MCP-native framing and portability.** We argue that MCP is not only an integration transport but a key enabler of framework-agnostic runtime verification.

Implementation and evaluation contributions (Part II):

- **OpenClaw + Cursor-capable EDAMAME instantiation.** A concrete deployment path over local, PSK-authenticated MCP with rollback-capable action handling and merged multi-agent reasoning-plane inputs. The quantitative benchmark in this draft uses OpenClaw-derived scenarios with explicit prebuilt-window behavioral-model slices published into EDAMAME, while Cursor is integrated through the raw-session observer contract rather than treated as a separate detection engine.
- **Trace-backed benchmark protocol.** Every verdict is tied to on-disk telemetry artifacts for auditability.
- **Empirical performance characterization.** Precision/recall, latency, and failure modes are measured on a live 50-scenario suite, including explicit designed-miss scenarios, and cross-checked with layered model-to-verdict validation.

Part I — Two-Plane Model

Part I is organized as follows: Section 2 positions this work in prior runtime security literature; Section 3 defines the two-plane model; Section 4 states threat assumptions and boundaries; and Section 5 gives concise properties of the decision rules as a bridge to implementation. Readers primarily interested in model

mechanics and applications may skip Section 2 and proceed directly to Section 3.

2. Background and Related Work

2.1 Attack Surface

Indirect prompt injection blurs data and instructions in LLM-integrated applications, enabling remote compromise via retrieved content (Greshake et al., 2023). Benchmarks such as InjecAgent show tool-integrated agents remain vulnerable to indirect prompt injection (Zhan et al., 2024).

MCP expands the attack surface by standardizing tool descriptors and invocations. Attacks embedded in tool metadata (tool poisoning, descriptor mutation) achieve attack success rates up to 72.8% on capable models (MCPTox; Luo et al., 2025). Mitigations include descriptor signing, semantic vetting (Jamshidi et al., 2025), and multi-stage neural detection pipelines achieving 96% accuracy on adversarial prompts (MCP-Guard; Chen et al., 2025). MCP's own specification documents authorization and security best practices, including confused deputy risks in proxying designs (MCP Spec, 2025). MCP Security Bench provides a 2,000-instance, 12-attack-type evaluation suite for systematically measuring tool-level defenses (Wang et al., 2025).

Goal drift – an agent's tendency to deviate from original objectives over extended execution – is now measurable in controlled evaluations. Zhou et al. (2025) show that all evaluated models exhibit some drift under autonomous operation, even when strong scaffolding substantially reduces it.

2.2 Runtime Defense Landscape

The 2025-2026 research cycle has produced a rich landscape of runtime defense systems. We organize them by their primary signal source.

Reasoning-plane defenses monitor the agent's internal state (prompts, chain of thought, tool call sequences) to detect compromise:

- AgentArmor (Gao et al., 2025) treats runtime traces as program objects, constructs CFG/DFG/PDG views, and applies type-checking to trust-boundary flows. It reports 97% defense success (3% attack success) on AgentDojo with modest utility impact.
- LlamaFirewall (Meta, 2025) provides three production guardrails: PromptGuard 2 (BERT-based jailbreak detection), Agent Alignment Checks (chain-of-thought auditing for prompt injection and goal misalignment), and CodeShield (online static analysis). It is deployed in production at Meta.
- AegisLLM (Li et al., 2025) uses cooperative multi-agent defense where orchestrator, deflector, responder, and evaluator agents collaborate for safe outputs, reporting a 51% improvement on jailbreak benchmarks without retraining.

- A2AS (Wallarm et al., 2025) proposes the BASIC security model: behavior certificates, authenticated prompts, security boundaries, in-context defenses, and codified policies. The initiative has broad industry sponsorship and is emerging as a framework-level governance model.

Hybrid defenses combine reasoning-state signals and runtime action signals:

- MI9 (Singh et al., 2025) is the first integrated runtime governance framework, combining goal-conditioned drift detection, FSM-based conformance engines, continuous authorization monitoring, and graduated containment. Reports 99.81% detection on synthetic scenarios.
- AgentSentinel (Hu et al., 2025) intercepts all sensitive operations within agent services, halting execution until a security audit correlating task context with system traces completes. Achieves 79.6% defense success across 60 attack scenarios in 6 categories.
- Pro2Guard (Zhang et al., 2025) learns Discrete-Time Markov Chains (DTMCs) from execution traces and proactively predicts unsafe states before they occur, with PAC-style correctness bounds. Reported defense success is 97.4% (2.6% unsafe outcomes).

Formal verification approaches provide provable safety properties:

- ShieldAgent (Xu et al., 2025) constructs safety policies as verifiable rule circuits and generates shielding plans with executable code for formal verification. Achieves 90.1% recall on safety violations.
- Agent-C (Kamath et al., 2025) enforces temporal safety constraints via SMT solving and constrained generation, achieving 100% conformance on leading-model evaluations.
- VET (Li et al., 2025) establishes verifiable execution traces via trusted hardware, cryptographic proofs, and notarized TLS transcripts with practical overhead under 3x for API calls.
- AgentGuard (Koohestani, 2025) provides runtime verification via probabilistic model checking and Dynamic Probabilistic Assurance for quantitative agent properties.

Chain-of-thought monitoring research reveals important limitations:

- Models can generate benign-sounding CoT explanations that mask misaligned intentions, causing monitors to fail (Thought Crime; Rogers et al., 2025).
- Agent awareness of monitoring is the single strongest evasion advantage – detection rates drop sharply when agents know they are monitored (Baker et al., 2025).
- Hybrid protocols scoring both reasoning and actions consistently surpass either alone, with 2x higher detection for subtle deception scenarios (CoT Red-Handed; Goldowsky-Dill et al., 2025).

2.3 Positioning

Table 2 summarizes the landscape. Our system provides MCP-native, multi-platform process-attributed telemetry integrated with intent correlation. The closest comparator is AgentSentinel, which also correlates task context with OS-level traces via eBPF/LSM. Key differentiators:

- **Multiplatform**: our observer (EDAMAME Posture) works across Linux, macOS, and Windows (Section 4.3), while AgentSentinel requires Linux eBPF/LSM. eBPF enhances attribution resolution on Linux but is not required.
- **MCP-native**: telemetry is exposed as standard MCP tools over Streamable HTTP with PSK authentication, usable by any MCP-compatible agent without custom integration.
- **Correlation vs. interception**: AgentSentinel halts execution at sensitive operations for synchronous audit; our system primarily correlates observed behavior after it appears in telemetry, with an optional proactive layer for MCP-gated actions. This trades immediate blocking for lower runtime friction and broader platform compatibility.
- **Latency trade-off (explicit).** Our measured 45.8 s median detection delay (p95: 53.6 s) is a practical limitation of reactive observation; AgentSentinel reduces this gap for intercepted operations but at the cost of inline interception and Linux/eBPF coupling.

Its primary limitations relative to the field are partial reasoning-plane coverage (transcript-only, no CoT introspection; Section 9) and absence of formal safety proofs.

System	Plane	MCP	Platforms	Eval. Scale
AgentArmor	Reasoning	No	Any	AgentDojo (97%)
LlamaFirewall	Reasoning	No	Any	Meta internal
AegisLLM	Reasoning	No	Any	Jailbreak bench.
A2AS BASIC	Reasoning	Yes	Any	Framework-level
MI9	Hybrid	No	Any	Synthetic (99.81%)
AgentSentinel	Hybrid	No	Linux (eBPF)	60 scen., 6 cat.
Pro2Guard	Hybrid	No	Any	Household + AV
ShieldAgent	Formal	No	Any	Web environments
Agent-C	Formal	No	Any	Multiple bench.
VET	Trace	No	Any	Black-box API
MCP-Guard	Metadata	Yes	Any	70,448 samples
This work	**Hybrid**	**Yes**	**Lin/Mac/Win**	**50 scen., 8 cat.**

A2AS BASIC is conceptual at the MCP-native layer. `This work` is hybrid because reasoning-plane sessions and system-plane telemetry are observed independently and merged only inside the observer-owned runtime.

Formal properties reported by prior systems: - A2AS (behavioral certificates) - MI9 (FSM conformance) - Pro2Guard (PAC-correctness) - ShieldAgent (rule circuits) - Agent-C (SMT temporal constraints) - VET (cryptographic proof) -

MCP-Guard (neural detector, reported 96%)

Deployed systems in this survey scope: - LlamaFirewall (Meta production) - A2AS (emerging standard) - This work (Lima VM) - All others are currently research-only artifacts.

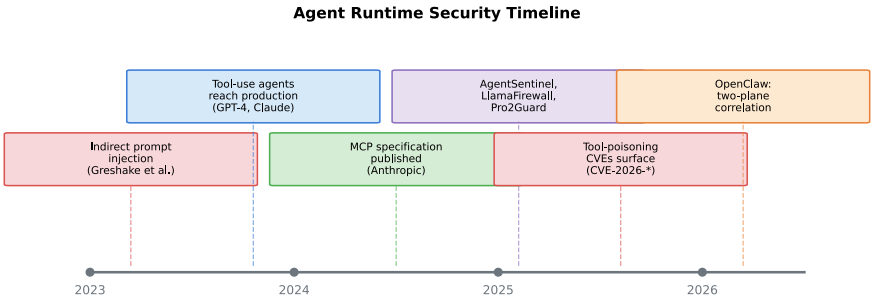

Figure 1: Agent runtime security timeline

No prior published defense, to our knowledge, exposes continual, process-attributed OS-level telemetry (eBPF sessions, parent process lineage, open file paths) as MCP tools and then demonstrates intent-telemetry correlation through that same interface.[1] Close neighbors (AgentSentinel, LlamaFirewall, MCP-Guard) intercept or audit tool calls but do not expose OS-level telemetry as a first-class MCP self-inspection surface. The trade-off is that system-plane monitoring is inherently reactive (Section 3.6) and reasoning-plane coverage is transcript-only, without CoT introspection (Section 9).

2.4 Reflection as Relevant Prior Art

The model is also related to reflection-oriented programming, where a program observes and reasons about its own behavior through a meta-level interface. Sobel and Friedman (1996) frame reflection as separating base-level execution from meta-level supervision. Our two-plane model adopts a similar separation: the reasoning plane executes and declares intent; the system plane independently observes effects and can challenge that declaration. We do not claim formal equivalence to classic reflection systems, but the analogy is useful for conceptual grounding.

[1] Survey scope (as of February 2026): arXiv, ACM DL, IEEE Xplore, and USENIX proceedings through February 2026 using keywords including "agent runtime monitoring", "MCP security", "tool-use telemetry", and "eBPF agent", plus manual review of MCP specifications and ecosystem tooling.

3. Two-Plane Architecture (Reasoning Plane and System Plane)

- **Reasoning plane:** what the agent declares it is doing and why.
- **System plane:** what the host actually does.

At a high level, the control loop is:

declare intent ->
derive expected effects ->
observe host behavior ->
correlate ->
compute deterministic evidence ->
apply LLM adjudication with critical-signal guardrails ->
emit final divergence verdict

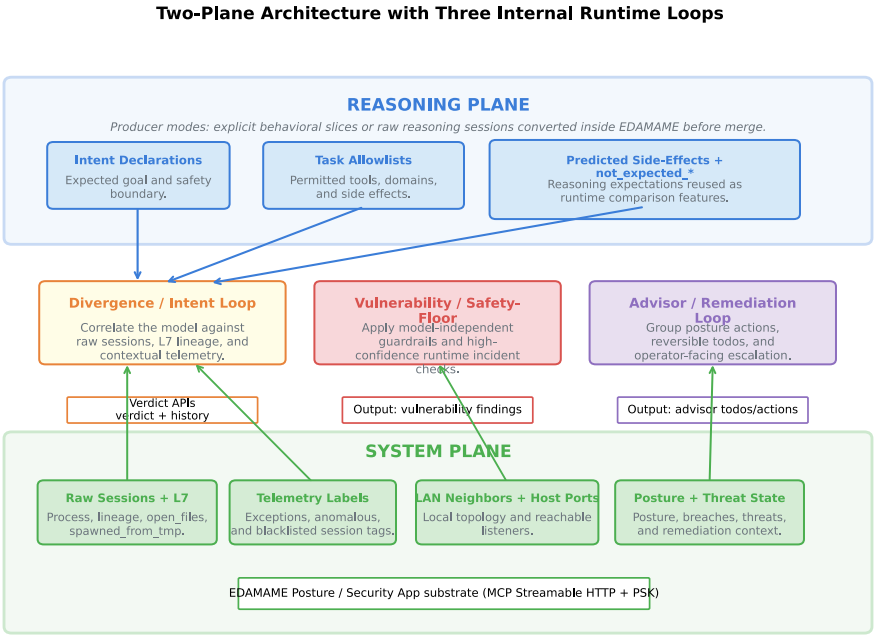

Figure 2: Two-plane architecture with reasoning-plane declarations and system-plane runtime loops.

The architecture maps reasoning-plane intent declarations to system-plane telemetry correlation, and emits divergence when observed behavior is not explained by declared intent. In the deployed system, the behavioral model consumed by the divergence engine can be a merged window assembled from multiple agent-tagged contributors (for example OpenClaw plus Cursor) rather than a single producer. Producer modes are additive: OpenClaw commonly

uses the prebuilt-window publication path, while workstation integrations such as Cursor can forward raw reasoning sessions so EDAMAME generates the contributor slice internally before merge.

The operational cadence is shared-timer-driven: extrapolator and divergence verdict-reader jobs run on offset timers, while EDAMAME evaluates three adjacent runtime loops internally. The timer structure is important to portability: it avoids dependence on framework-specific synchronous hooks and keeps behavior consistent across deployments. We considered two alternatives – interception-first synchronous gating (lower delay but higher runtime friction and tighter platform coupling) and a single merged polling loop (simpler but weaker separation between model publication and verdict consumption). The shared-timer split gave the best operator clarity while preserving framework-agnostic deployment.

The paper's model claim remains two-plane correlation. The three-loop split is an implementation decomposition of the system plane in the reference instantiation:

1. **Divergence / intent loop.** Correlates the merged behavioral model against raw process-attributed telemetry and emits intent-relative evidence when observations are not explained by declared behavior.
2. **Vulnerability / safety-floor loop.** Evaluates model-independent guardrail findings and runtime incident checks (for example `/tmp` lineage, credential-file-plus-egress, exposed listeners, and other vulnerability detectors), including cases where the behavioral model is missing or stale.
3. **Advisor / remediation loop.** Groups posture, breach, LAN, and high-confidence telemetry findings into operator-facing todos and reversible actions.

This separation is deliberate: telemetry confidence labels such as `anomalous` or `blacklisted` are useful operational signals, but they are not treated as intent-relative divergence evidence by themselves. Scenarios are judged from engine verdicts read via `get_divergence_verdict`; the final operator verdict may be LLM-adjudicated, but deterministic safety-floor and vulnerability-loop findings remain non-downgradable.

3.1 Reasoning-Plane Inputs

The reasoning plane provides structured declarations:

- intent statement (task-scoped)
- tool plan and rationale
- predicted side effects
- task-conditioned allowlists (processes/domains/paths)

Session-transcript observation (reference implementation, Part II). This detail matters because it changes the trust model of the reasoning plane. In the reference implementation, the reasoning plane is not self-declared. For

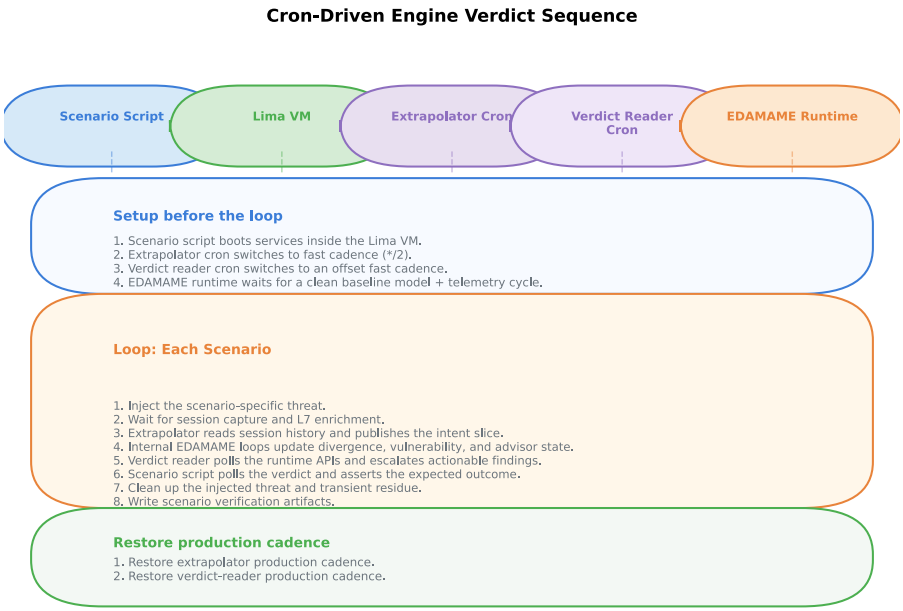

Figure 3: Shared-timer cadence for extrapolator publication, internal runtime loops, and verdict consumption.

OpenClaw, a separate monitor session reads other sessions' tool calls and reasoning via the platform's `sessions_list` and `sessions_history` APIs. For Cursor, a parallel publisher emits the same behavioral-window schema with its own `agent_type` and `agent_instance_id`. EDAMAME stores these slices, preserves contributor attribution, and merges them before correlation. This means an agent cannot retroactively edit or suppress evidence of its tool calls when the transcript or observer surface is platform-controlled. The reasoning plane therefore shifts from trust-dependent self-declaration to an independently verifiable record, though transcript integrity still depends on the platform (see Section 9, Limitations).

3.2 System-Plane Observations

The system plane consumes independent host telemetry:

- network sessions with **Layer-7 process enrichment** (binary, parent lineage, open files, command line, resource usage)
- telemetry confidence labels (`exceptions`, `anomalous`, `blacklisted`)
- host LAN visibility and local open-port observations
- posture score deltas and threat-model outputs

These telemetry labels are organized as a strict confidence hierarchy:

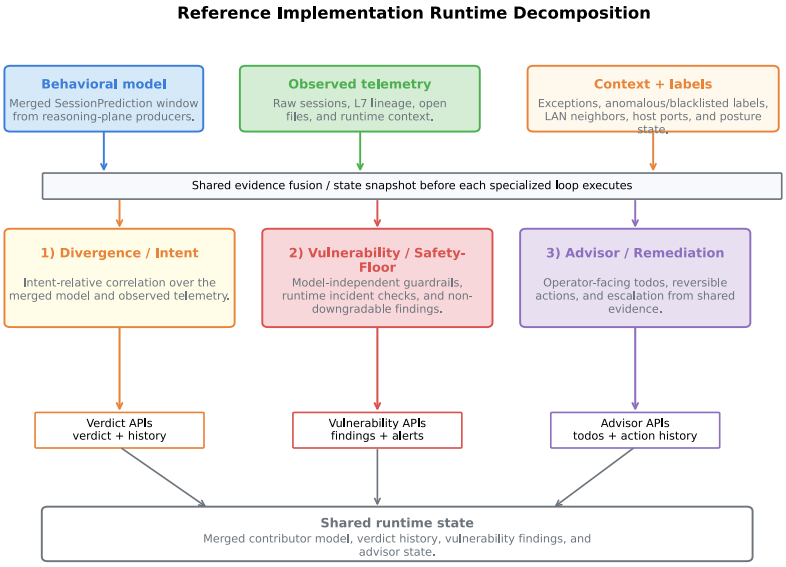

Figure 4: Reference runtime decomposition of the three adjacent loops inside the system plane.

```
sessions -> exceptions -> anomalous -> blacklisted
```

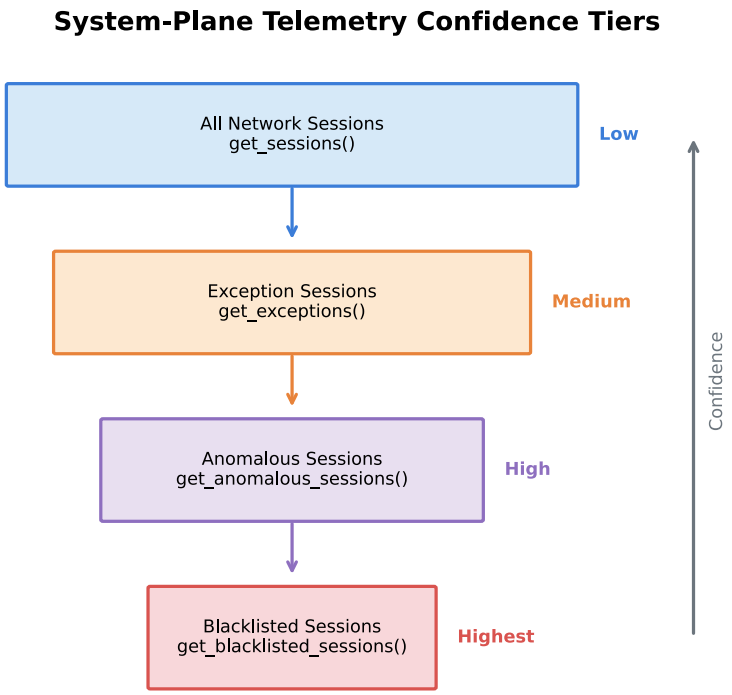

Figure 5: Signal hierarchy with escalating detection confidence

The hierarchy is part of the model as a system-plane evidence-prioritization mechanism, not merely an EDAMAME implementation detail: it separates "did anything happen?" from "how strongly should we trust that this is malicious?" and allows conservative policy tuning at lower tiers. In the reference implementation, however, those labels do not replace intent correlation. The divergence / intent loop reasons primarily over raw sessions plus Layer-7, LAN, and open-port context; the vulnerability / safety-floor loop and the advisor / remediation loop consume `anomalous` and `blacklisted` labels as high-confidence operational signals.

3.3 Why Process Attribution Is Model-Critical

The two-plane model requires process attribution to remain meaningful:

- without process attribution, an allowlist collapses to destination-only checks, so benign and malicious programs contacting the same endpoint become indistinguishable
- parent-process lineage is required to detect spawn-chain manipulation, where an approved binary is launched through an untrusted wrapper chain
- open-file context is required for credential-adjacent detection, because outbound traffic alone cannot show whether sensitive files were read just before egress

This requirement is independent of implementation. Any observer that cannot attribute connections to concrete processes cannot fully realize the model.

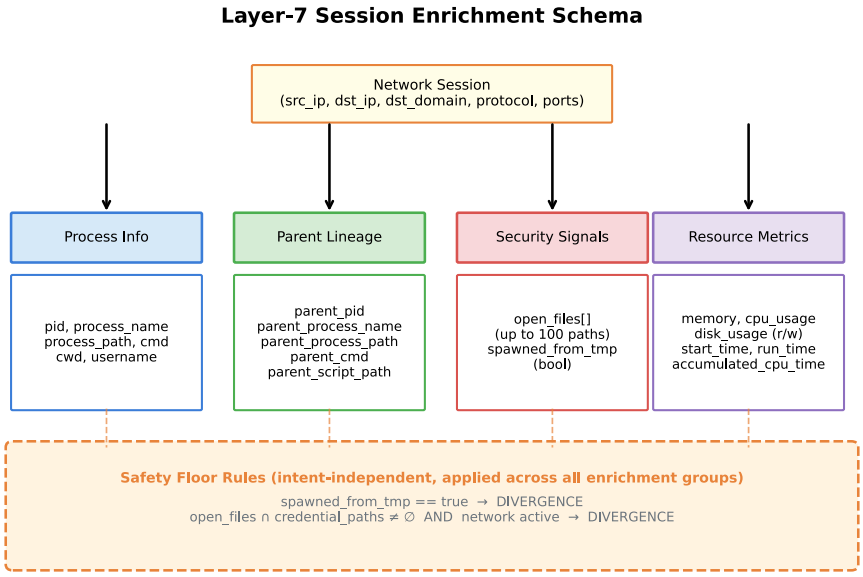

Figure 6: Layer-7 Session Enrichment Schema.

Each session record is enriched with process identity, parent lineage, security signals (including `open_files` and `spawned_from_tmp`), and resource metrics. In this schema, safety-floor rules apply across the full evidence bundle, not only to a single branch.

3.4 Why MCP Matters to the Model

MCP is not only a transport convenience. It makes the model operationally portable:

1. **Structured semantics.** Typed responses reduce parsing ambiguity and let correlation logic consume machine-readable evidence directly.

2. **Transactional actions.** Execute/rollback patterns make pre-action and post-action checks composable.
3. **Framework portability.** MCP-facing tools can be consumed by OpenClaw, Cursor, other coding agents, and other MCP-compatible clients with minimal glue code. In the deployed contract, each producer tags its slice with agent identity and EDAMAME performs the merge centrally.

As positioned in Section 2.3, exposing continual process-attributed host telemetry as first-class MCP tools is central to this instantiation.

3.5 Multiplatform Requirement and Instantiation

Multiplatform operation is not cosmetic. Agent workflows already span Linux servers, macOS developer hosts, and Windows enterprise endpoints. This is especially true for coding agents operating across local repos, CI runners, and production deployment environments.

Accordingly, the model requires attribution beyond Linux-only eBPF paths. Our current implementation uses:

- Linux eBPF where available
- native macOS/Windows process APIs
- a socket-matching fallback when kernel telemetry is unavailable

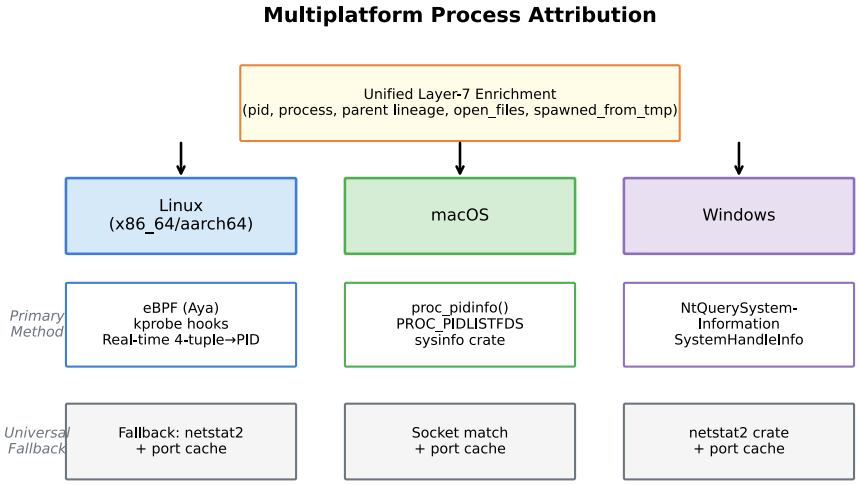

Figure 7: Multiplatform Process Attribution: Linux uses eBPF kprobes for real-time attribution, macOS uses proc_pidinfo, and Windows uses NtQuerySystemInformation. All platforms share a universal fallback via netstat2 socket matching with port cache.

3.6 Observation Delay Is a Constraint, Not a Contribution

Telemetry is delayed relative to action execution. We therefore use a settling schedule (e.g., 15 s -> 30 s -> 60 s) and stop when snapshots stabilize. This is an operational accommodation, not a conceptual novelty claim.

3.7 Two Example Rules (Simple and Realistic)

We present two example rules: one intentionally simple and one more realistic. The rules themselves are not claimed as contributions; they are concrete examples of model use and an introduction to the rule properties discussed in Section 5.

- **Example Rule A (simple, reactive):** intent-vs-observation correlation after telemetry arrives.
- **Example Rule B (more realistic, proactive+reactive):** pre-action blocking for MCP-gated operations plus Rule A for residual detection.

Example Rule A: Reactive Intent-vs-Behavior Correlation

```
process_allowed = (evidence_process in allowed_processes)
domain_allowed  = (target_ip in allowed_domains)
explained       = process_allowed AND domain_allowed

guardrail_signal = (safety_floor_findings > 0)
                OR (vulnerability_findings > 0)

intent_mismatch  = (session_matches > 0 AND explained == false)
                OR (credential_access AND session_matches > 0)

divergence = guardrail_signal OR intent_mismatch
```

Example Rule B: Pre-Action Guard Rule

```
action = proposed tool call (name, arguments)
intent = current declared intent

spawns_from_tmp     = action.working_dir starts_with /tmp/ | /var/tmp/ | /dev/shm/
targets_credentials = action.arguments reference sensitive file paths
targets_blacklisted = action.arguments contain known-blacklisted endpoints

process_in_allowlist = action.tool_binary in intent.allowed_processes
target_in_allowlist  = action.target_endpoint in intent.allowed_domains

static_violation = spawns_from_tmp OR targets_credentials OR targets_blacklisted
intent_violation = NOT (process_in_allowlist AND target_in_allowlist)

block = static_violation OR intent_violation
```

In this draft, "sensitive credential paths" is a configurable static set (for example `~/.ssh/*`, kube configs, cloud credential stores, browser-token stores, and CI token files) defined by deployment policy rather than inferred at runtime.

Rule B does not replace Rule A: actions bypassing MCP still require reactive system-plane correlation.

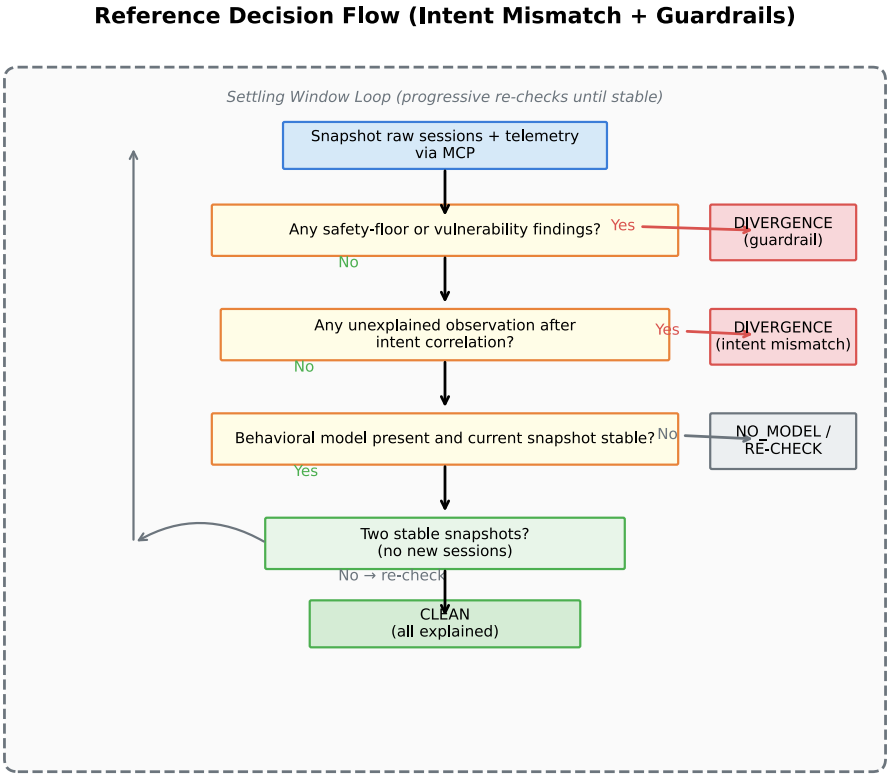

Figure 8: Detection Decision Flow (Example Rule A).

In this flow, sessions are checked against blacklists, vulnerability-detection checks, and intent-conditioned allowlists within a settling-window loop. Divergence is raised at the first failing check.

Model-level multi-signal scenarios (network + identity + LAN + posture) are provided in Appendix A to keep the main narrative focused.

4. Threat Model, Assumptions, and Boundaries

We focus on runtime compromise modes that evade static checks:

- indirect prompt injection leading to unsafe tool use
- tool/MCP poisoning and capability misuse
- unexplained outbound communications during ostensibly read-only tasks
- credential-path and config-adjacent anomalies
- cross-session behavioral drift and memory poisoning

4.1 Assumptions and Why They Are Reasonable

Assumptions

- The observer is co-located with the runtime host.
- Baseline S1/S2 controls are not skipped.
- High-risk actions can require human confirmation when enabled.

Rationale

- Co-location is necessary for process attribution and parent-lineage fidelity.
- If S1/S2 are absent, runtime detection is overloaded with preventable risk.
- Human confirmation for destructive actions is standard in production ops.

Out of scope (with rationale)

- Formal properties under arbitrarily compromised reasoning planes (requires stronger mechanisms such as formal methods or trusted execution).
- Internet-scale adaptive adversarial campaigns (requires different evaluation infrastructure than the controlled benchmark used here).

4.2 Reasoning-Plane Integrity

The key question raised by reviewers is valid: if an attacker can control intent declarations, can they simply declare harmful intent and pass correlation? The answer depends on which failure case occurred:

- **Case A - Tool misuse within declared benign intent.** The model is strong. Divergence appears because observed behavior exceeds declared intent.
- **Case B - Correctly declared harmful intent.** Correlation alone is insufficient if policy allows that action. Additional governance controls (environment-aware allowlists, posture-delta thresholds, change-control invariants) are required.
- **Case C - Missing blast-radius guardrails.** If destructive production actions are broadly authorized, the root cause is authorization architecture, not telemetry correlation. The two-plane model complements S1/S2; it does not replace IAM scoping, segmentation, and approval workflows.

To raise attacker cost in Cases A/B, we apply two mitigations.

Intent-independent critical signals. The following triggers cannot be downgraded by LLM adjudication or intent declarations:

1. blacklisted destination access

2. CVE-inspired vulnerability-detection failures (gateway binding, token exfiltration, skill supply chain, sandbox exploitation)

Model-correlated signals (unexplained sessions, credential access concurrent with outbound traffic, anomalous network patterns) are promoted as divergence evidence and adjudicated by the LLM. The LLM always makes the final verdict decision, but critical signals act as non-downgradable guardrails.

Intent-declaration anomaly tracking. Motivation: compromised agents often "prepare the ground" by gradually broadening intent language before executing harmful actions. We therefore track:

1. allowlist budget expansion across redeclarations
2. abrupt intent-category shifts
3. redeclaration churn (rate spikes)

This signal is currently specified at the model level; its marginal contribution is not isolated in the Section 7 confusion-matrix metrics.

These mitigations do not remove the trust boundary entirely, but they force an attacker to simultaneously evade both declaration-level and system-level checks.

4.3 Observer Integrity and Evasion Boundaries

The model assumes a trustworthy observer process. Under kernel/root compromise, the observer itself can be tampered with. Potential hardening paths include watchdog heartbeat, cryptographic attestation, and stronger isolation.

Primary residual evasion classes are:

- **Living-off-the-land** (allowed process + allowed destination)
- **timing attacks** (harm before first informative snapshot)
- **observer evasion/tampering**
- **credential-less exfiltration** (transient file access not captured in time)

The most operationally important combined risk is living-off-the-land plus timing. This combination is dangerous because behavior can remain syntactically "allowed" while executing before the most informative telemetry fields are available. Near-term mitigations are shorter enrichment intervals, sticky process-context retention, stricter defaults for high-risk tasks, and proactive Rule B gating for MCP-mediated operations. We detail mitigation priorities and benchmark blind spots in Section 9.

4.4 State Isolation and Observer Independence

A subtle but critical attack surface in agentic frameworks is *shared mutable state*. Many frameworks persist inter-session context – behavioral models, verdicts, operational memory – in flat files within the agent's workspace (e.g., `MEMORY.md`, `SOUL.md`). These files are readable and writable by any process on the host. This creates multiple attack vectors:

1. **Model poisoning.** A compromised process writes false behavioral predictions into the shared memory file, causing the detection layer to whitelist malicious traffic patterns as "expected."
2. **Verdict tampering.** A compromised process overwrites the stored verdict (e.g., writing `CLEAN` over a `DIVERGENCE` determination), suppressing alerts before they reach human operators.
3. **Self-covering reasoning plane.** A compromised agent can edit its own memory to erase evidence of divergent behavior, defeating any retrospective audit.

These attacks are particularly dangerous because they require no privilege escalation – only filesystem write access to a user-owned directory.

Our architecture mitigates this by *internalizing detection state*. The divergence engine runs inside the EDAMAME host runtime: the EDAMAME Posture daemon in the reference benchmark deployment, and the EDAMAME Security app on workstation-hosted Cursor paths. Behavioral models, verdict history, and engine configuration live in process memory and are accessible only through authenticated MCP endpoints (pre-shared key). Multiple reasoning-plane producers can push agent-tagged slices via `upsert_behavioral_model`; EDAMAME merges those slices into one observer-owned correlation window while retaining contributor attribution metadata. The reasoning-plane agent can contribute predictions, but cannot read or modify verdicts, engine state, or critical-signal rules.

This separation provides three concrete assurances:

- **Unilateral tamper resistance.** Compromising the reasoning plane alone is insufficient to suppress detection. The observer's state is unreachable from the agent's process or filesystem.
- **Independent alerting.** The observer-owned notification pipeline (Slack, Telegram, webhook) is shared across agentic and divergence/vulnerability outputs, not owned by the agent. Even a fully compromised reasoning plane cannot intercept or cancel outbound alerts.
- **Critical-signal invariance.** Intent-independent rules (Section 4.2) are evaluated inside the observer. No external input can downgrade a critical-signal finding.

The extrapolator skill retains OpenClaw's memory system for its own operational checkpoint (a cursor tracking the last-processed session). This is acceptable: the checkpoint is not security state. Corrupting it causes the extrapolator to reprocess sessions but cannot influence EDAMAME's verdicts.

5. Properties of the Decision Rules

This section can be read as a short extension of Section 4 (Threat Model): it states properties of the deterministic decision rules (Example Rules A-B) that hold by construction. These are not deep theorems; they are simple, auditable

invariants meant to make system behavior transparent. We explicitly distinguish these from stronger formal properties (PAC bounds, temporal SMT constraints, cryptographic proofs) provided by systems such as Pro2Guard, Agent-C, and VET (Section 2.2). Our contribution is practical, not formal.

5.1 Definitions

For each step t:

- I_t: declared intent (reasoning plane)
- C_t: context constraints derived from I_t (e.g., allowlists)
- O_t: system-plane observation snapshot after settling
- K(O_t): critical-signal predicate (true if blacklisted traffic or vulnerability-detection failure)
- E(I_t, C_t, O_t): "explained" predicate (true if O_t is consistent with intent)

The two-plane alert predicate is:

$$D(I_t, C_t, O_t) = K(O_t) \vee (O_t \neq \emptyset \wedge \neg E(I_t, C_t, O_t))$$

5.2 Property 1 (False-Positive Monotonicity)

For any anomaly predicate A over observations, let D_gated = A AND (NOT E) and D_critical = K OR D_gated. Then for benign runs where K does not fire:

```
P(D_critical = 1 | benign, K=0) = P(D_gated = 1 | benign) <= P(A = 1 | benign)
```

This follows from set inclusion: {D_gated=1} implies {A=1}. Intent-gating cannot increase false positives relative to anomaly-only alerting when no critical signal is triggered.

Remark (critical-signal false positives). Critical signals *can* produce false positives for benign operations that coincidentally match (e.g., a legitimate connection to a recently-blacklisted destination).

5.3 Property 2 (Settling Window and Miss Rate)

Let L be the time until an action's effects become observable. With maximum wait W: P(miss) = P(L > W). Progressive re-checks (sampling at t1 < t2 < … < W) do not change this bound but reduce the expected detection latency when L < W.

5.4 Property 3 (Example Rule A Correctness Under Complete Telemetry)

Assume the evidence bundle is complete: `session_matches > 0` iff the endpoint was contacted by `evidence_process` after the watermark. Example Rule A

returns `divergence=true` iff the observed egress is not authorized by intent OR a critical signal fires. This follows directly from the algorithm's structure and is not a claim about real-world completeness. In practice, this assumption is violated by settling windows and refresh delays (Section 3.6, Section 9.1), so Property 3 should be read as a rule-level sanity property, not an empirical coverage claim.

5.5 Property 4 (Pre-Action Blocking is Conservative)

Example Rule B blocks on `static_violation OR intent_violation`. Since both predicates are computed from the proposed action and current intent (no telemetry), blocking is immediate (zero latency). The cost is over-blocking: legitimate actions that happen to match a floor pattern (e.g., a build tool that writes to `/tmp/`) require human confirmation.

5.6 Scope of These Properties

These properties characterize the decision rules under idealized assumptions (complete telemetry, correct allowlists, honest intent for Property 1). They do not provide:

- Safety properties under partial telemetry (missed sessions, eBPF gaps)
- Bounds on evasion probability (see Section 4.3)
- Temporal safety properties (no ordering constraints on actions)
- Probabilistic properties on detection rates (the Wilson CI in Section 7.4 is empirical, not analytical)

Systems that require these stronger properties should consider Pro2Guard (PAC bounds), Agent-C (temporal SMT), or VET (cryptographic attestation).

Part I takeaway: the model provides a portable and auditable way to correlate declared intent with host-observed behavior, but by itself it does not provide zero-latency interception, complete living-off-the-land coverage, or formal safety proofs. Part II therefore focuses on implementation feasibility and measured operational behavior.

Part II — Implementation and Evaluation

Part II demonstrates feasibility and relevance: Section 6 describes our implementation, Section 7 reports evaluation results, Section 8 documents reproducibility assets, Section 9 states limitations and near-term work, and Section 10 discusses ethics and safe use.

6. Implementation (OpenClaw + EDAMAME Posture)

Our implementation uses a shared-timer architecture with one OpenClaw reasoning skill, EDAMAME-internal detection/remediation loops, and an internal EDAMAME divergence engine:

Two-Plane Detection (cooperating timer jobs):

- `skill/edamame-extrapolator/` – Reads raw session history via `sessions_list(activeMinutes=15)` and `sessions_history`, distills a compact behavioral model, and publishes it via `upsert_behavioral_model`. Runs every 2-5 minutes. Uses per-session `message_count` checkpoints to implement a sliding window and avoid re-analysis.
- **Internal divergence engine** (inside EDAMAME Posture) – Implements a hybrid Rust-deterministic + LLM-adjudication architecture. Runs every ~5 minutes by default (configurable). Each tick proceeds in three stages:
  1. **Deterministic evidence stage** (Rust, `divergence_engine.rs`): Stateless pure functions correlate behavioral model predictions against live telemetry (sessions, anomalous, blacklisted, exceptions) and run CVE-inspired vulnerability-detection checks (gateway binding via `ourselves.open_ports`, token exfiltration, skill supply chain, sandbox exploitation). Produces a baseline `DivergenceVerdict` with typed evidence (CRITICAL/HIGH/MEDIUM severity).
  2. **LLM adjudication stage**: The deterministic evidence bundle is submitted to the configured LLM provider for final verdict decision and operator-facing alert synthesis. The LLM receives the full evidence context and returns a structured `LlmDivergenceDecision` (verdict, confidence, reasoning) plus an optional `LlmHumanAlert` (title, body, severity).

3. **Guardrail reconciliation** (`reconcile_with_guardrails`): Critical signals – vulnerability-detection findings and blacklisted-traffic evidence – are non-downgradable. If the LLM attempts to override a CRITICAL deterministic finding, the guardrail forces DIVERGENCE with provenance `GUARDRAIL_FORCED_DETERMINISTIC`. Otherwise, the LLM verdict is accepted with provenance `LLM_CONFIRMED` or `LLM_OVERRULED_SOFT_SIGNALS`. Each verdict records full provenance: `deterministic_verdict`, `decision_source`, optional `llm_decision`, and optional `human_alert`. If the behavioral model is stale (>20 min), produces a STALE verdict. Verdicts are accessible via `get_divergence_verdict` MCP tool.

Posture Management (EDAMAME-internal loop):

- EDAMAME posture/remediation loop (internal) – Reviews advisor todos and executes safe fixes through the native agentic pipeline. In OpenClaw, `skill/edamame-posture/` acts as a thin MCP facade for invoking these capabilities; scheduled remediation ownership remains inside EDAMAME.

Supporting components:

- OpenClaw plugin bridging to MCP: `extensions/edamame-mcp/`
- Live benchmark harness: `tests/benchmark/run_live_suite.py`
- Metric summarizer: `tests/benchmark/summarize_results.sh`
- Canonical claim binding: `docs/CLAIM_ARTIFACT_INDEX.md`

Behavioral-model and verdict state are internal to the EDAMAME divergence engine. OpenClaw memory remains optional operational checkpointing for the extrapolator cursor and is not authoritative detection state.

6.1 MCP Integration Path

EDAMAME Posture exposes a local, PSK-authenticated MCP endpoint via Streamable HTTP (`http://127.0.0.1:3000/mcp`). OpenClaw calls EDAMAME MCP tools through a local OpenClaw plugin (`extensions/edamame-mcp`) that implements a minimal HTTP MCP client and exposes EDAMAME tools as native OpenClaw tools:

OpenClaw (native tool call) ->
`edamame-mcp` plugin ->
EDAMAME MCP (HTTP) ->
tool result

MCP Integration Notes Section 3.4 explains why MCP is model-relevant. At implementation level, the key point is that OpenClaw can consume EDAMAME as native tools (typed JSON over local PSK-authenticated HTTP), so correlation logic can run without shell parsing or privileged subprocess glue.

6.2 Trace-Backed Evidence Bundles

The live benchmark suite records:

- baseline and post-action telemetry snapshots
- inject logs with VM-side timestamps
- MCP call outputs (todos, action history)
- a run manifest with scenario set hash and git SHA

These artifacts enable independent inspection of every verdict.

6.3 Claim Gating and Validation Maturity Tiers

To avoid confusion between runtime **evidence bundles** (system telemetry used by the divergence engine) and paper-level **validation status** (what this manuscript has empirically established), we use the following maturity tiers:

- T0: implemented
- T1: demonstrated in integration tests
- T2: measured in trace-backed benchmark artifacts
- T3: independently replicated

Automation blocks unsupported strong-claim language unless minimum tier floors are met. Published metric claims are anchored to a single manifest/summary lineage in `docs/CLAIM_ARTIFACT_INDEX.md`, so run IDs, hashes, and benchmark modes remain auditable.

6.4 Signal Coverage vs Validation Tier (What Is Empirically Supported)

The system targets multi-signal attacks, but empirical support is non-uniform. Table 1 summarizes current status:

Channel	Tier	Notes
Network egress (sessions, anomalous, blacklisted)	T2	Trace-backed live suite (paper metrics)
Process lineage (parent_pid, parent_name, parent_path)	T1	Layer-7 enrichment integration-tested via MCP
Open file access (per-process file descriptors)	T1	Credential-theft detection via open_files
Exceptions / policy violations	T1	Sanity-checked, no dedicated T2 protocol
MCP wiring (advisor_get_todos, action_history)	T1	OpenClaw -> plugin -> MCP path verified
Posture scoring (get_score, advisor_get_todos)	T1	Triage/demo flows; not yet T2-measured
Vulnerability detection (config/env safety scans)	T1	Unsafe configuration plus host-open-port gateway exposure checks
Cross-session drift (memory poisoning)	T0/T1	Partially exercised; not in T2 protocol

6.5 Operational Context: Posture Scoring and Compliance

Beyond individual network signals, EDAMAME Posture continually computes a composite security score (0-100) with subcategory breakdowns (network, system integrity, system services, applications, credentials) and maps the endpoint state against compliance frameworks (CIS Benchmark Level 1/2, ISO 27001/2, SOC 2). This provides two advantages for the two-plane model:

1. **Baseline context.** Endpoint hardening level (e.g., 90/100 vs 45/100) changes risk posture and response aggressiveness.
2. **Score-delta detection.** A posture drop during a task (e.g., 65 to 50) is an independent system-plane signal, even when individual sessions look benign.

These capabilities are exercised in the extrapolator/internal divergence engine flow and demo path (T1). Their incremental impact on T2 live detection accuracy remains future work. Even at current maturity, this context already

improves operator triage and supports safer action review in T1-validated workflows.

7. Evaluation: The BadAgentUse Benchmark

We introduce **BadAgentUse**, an open benchmark for evaluating agentic-system monitors against real endpoint telemetry. Unlike prior benchmarks that simulate network traces or rely on sandboxed environments, BadAgentUse injects live traffic into a production-grade Linux VM and evaluates detection against actual EDAMAME Posture session captures.

7.1 Setup

- **Environment:** Lima VM (`openclaw-security`), Ubuntu 24.04, 4 vCPU / 4 GiB
- **Observer:** EDAMAME Posture daemon (eBPF-backed on aarch64 Linux) with MCP Streamable HTTP endpoint on port 3000
- **Reasoning-plane path:** In production, OpenClaw publishes behavioral-model slices through the `edamame-extrapolator` skill, and Cursor publishes through the same EDAMAME contributor contract. For the quantitative BadAgentUse runs reported here, however, the harness injects scenario-specific behavioral-model slices directly via `upsert_behavioral_model` after deriving them from the scenario contract and live baseline telemetry. This isolates the downstream divergence engine from cron cadence and LLM nondeterminism. End-to-end extrapolator behavior is validated separately by integration and E2E tests rather than by the main aggregate metric loop.
- **Protocol:** 50 versioned JSON scenario files in `tests/benchmark/live-scenarios/`
- **Iterations:** configurable via `--iterations`; the current canonical artifact lineage is a partial live sweep rather than a uniform full grid
- **Seeds:** 12 distinct seeds in the current canonical live evidence set, with deterministic target-IP rotation
- **Trial unit:** one scenario evaluated per seed-iteration pair. The confusion matrix (Section 7.4) reports n=273 valid runs across 12 seeds, with 172 skipped runs preserved in the readiness scorecard. Each iteration varies seeds and settling behavior; all iteration traces remain in artifacts.

Protocol Parameters

Parameter	Value	Notes
Settling schedule	15 s → 30 s → 60 s	Exponential back-off; max wait W = 60 s
Stability criterion	2 consecutive stable snapshots	Snapshot = sessions + Layer-7 + posture unchanged

Parameter	Value	Notes
Layer-7 full refresh	300 s (5 min)	Process metadata, parent lineage, open files
Sensitive-file scan	30 s	Credential paths only; sticky retention
Anomaly-score threshold	0 (any non-zero triggers)	Conservative; tunable per deployment
Posture-delta threshold	$\geq$ 1 point drop	Monitoring signal (not hard-gate)
Blacklist sources	FireHOL Level 1 + custom	Pinned at VM provision time
Allowlist semantics	Exact domain OR IP; no wildcards	AND logic: process AND destination must match
MCP endpoint	`127.0.0.1:3000`, PSK-authenticated	Localhost-only by default
LLM (attack detection)	`openai/gpt-5.1`	Deterministic tiers do not use LLM; compatible Azure-hosted OpenAI Responses models are supported as alternate providers

7.2 Scenario Design (Versioned Categories)

In BadAgentUse, a **scenario** is a versioned JSON test contract that specifies: (1) declared intent context, (2) injected runtime action or traffic pattern, (3) expected enforcement label (`DIVERGENCE`, `CLEAN`, or `DESIGNED_MISS`), and (4) the telemetry checks needed for scoring.

BadAgentUse covers eight categories spanning five signal families:

Cat.	N	Signal	Description
A. Network-Intent	17	Network	Undeclared egress, partial-allowlist AND, lateral movement
B. Process Lineage	6	Layer-7	/tmp/ scripts, interpreter wrap, nested chains
C. Credential Theft	7	Layer-7 files	SSH, kube, gnupg, browser cookies, combined exfil

Cat.	N	Signal	Description
D. CVE Vulnerability Detection	6	Layer-7+allow	gateway_binding / token_exfil / credential overrides
E. Posture & Policy	–	Posture	Score, policy, threats, HIBP (integration checks)†
F. LAN Awareness	–	LAN	Device discovery, scan, anomaly APIs (integration checks)†
G. Adversarial	8	Mixed	LotL, timing, burst, path substitution
H. Noisy Benign	6	Network	Clean traffic, DNS, NTP, auto-update

† Categories E and F are validated via dedicated integration scripts (Section 7.7), not as JSON benchmark scenarios. The 50 benchmark scenarios span categories A–D, G, H.

Enforcement labels. Each scenario carries an `expected_enforcement` field: `DIVERGENCE` (39 scenarios), `CLEAN` (8 scenarios), or `DESIGNED_MISS` (3 scenarios). The `DESIGNED_MISS` label explicitly marks living-off-the-land attacks that Example Rule A cannot detect by construction (Section 4.3).

7.3 Metrics

We adopt the metric framework from AgentSentinel [Hu et al., 2025]:

- **DSR** (Detection Success Rate) = TP / (TP + FN) – equivalent to recall for attack scenarios
- **FPR** (False Positive Rate) = FP / (FP + TN) – rate of benign traffic incorrectly flagged
- **FNR** (False Negative Rate) = FN / (FN + TP) – attacks that evaded detection
- **ASR** (Attacker Success Rate) = FNR – attacks that succeeded despite monitoring
- **FT** (False Trigger) = FPR – identical to FPR for symmetry
- **TTD** (Time-to-Detect) = median and p95 detection latency (ms), computed from injection timestamp to first evidence capture
- **Wilson CI95** on DSR and FPR for statistical significance

All metrics are computed from trace-backed NDJSON results by `summarize_results.sh`, which also produces per-category and per-seed breakdowns for stability analysis.

7.4 Statistical Confidence Reporting (Wilson Interval)

To bound measured proportions (precision/recall), we use the Wilson score interval (Wilson, 1927). For `k` successes out of `n` trials and `z=1.96` (95% confidence):

```
p_hat = k/n
den   = 1 + z^2/n
center= (p_hat + z^2/(2n)) / den
half  = z * sqrt(p_hat*(1-p_hat)/n + z^2/(4n^2)) / den
CI95  = [center - half, center + half]
```

In our current canonical live evidence set:

	Predicted DIVERGENCE	Predicted CLEAN
Ground-truth attack	TP = 225	FN = 3
Ground-truth benign	FP = 36	TN = 9

Precision = 225/261 = 0.862 and recall = 225/228 = 0.987, yielding CI95 [0.815, 0.899] and [0.962, 0.996], respectively (n = 273 valid runs, 12 seeds).

The 3 FN are highly concentrated in two stress buckets: 2 occur in adversarial-evasion behavior, with the remaining miss in intent-proof-floor timing. Likewise, 22 of the 36 FP arise from noisy-benign traffic, with the remaining 14 in the broader network-intent-correlation bucket. The aggregate result therefore reflects localized boundary conditions rather than uniform degradation across categories, and points directly to two remaining needs: stronger benign-traffic baselining and better coverage of living-off-the-land behavior.

7.5 Results

Current Canonical Live Evidence Set *Results below come from the canonical evidence set in* ***`artifacts/live-paper-summary.json`*** *(50 planned scenarios; 273 valid runs across 12 seeds, with 172 skipped runs retained in the readiness scorecard). These aggregate metrics are from the OpenClaw-derived live benchmark population under the explicit-slice producer mode (`upsert_behavioral_model`) into EDAMAME; Cursor uses the raw-session producer mode (`upsert_behavioral_model_from_raw_sessions`) but is not a separate benchmark cohort in this draft.*

Claim-binding metadata (canonical evidence lineage):

- `run_id`: `run-live-20260304T234441Z-live`
- `git_sha`: `473465c`
- `scenario_set_version`: `22cb4a6cbddf88154435207c279143f7245de94dbc47c5803ca349eb10f3de1d`
- `mode`: `live`
- `benchmark_mode`: `live`

Metric	Value
Total runs	273
Precision	0.8620689655172413
Recall	0.9868421052631579
Precision CI95	[0.8149560047336936, 0.898678107127887]
Recall CI95	[0.9620350252457207, 0.9955153022192381]
Median latency (ms)	45810
p95 latency (ms)	53582
Rollback reliability	1
Stability (precision stddev)	0.10530453269195077
Stability (recall stddev)	0.07229988054812211

Readiness note. The current committed lineage remains below the repo's publication gate: `artifacts/arxiv-readiness-scorecard.json` is still `NO_GO` because the precision CI95 lower bound is 0.815 and the skipped-run ratio is 38.7%. We keep these values in the draft because they are the currently reproducible canonical artifacts on this branch.

Per-Category Detection

Category	Run N	DSR	FPR	Notes
Network intent correlation	68	1.000	0.700	undeclared egress is covered well, but benign allowance ambiguity still produces many false triggers
Process lineage verification	45	1.000	–	spawned_from_tmp, wrapper lineage, nested chains
Credential theft detection	36	1.000	–	SSH, kube, gnupg, browser-cookie access patterns
Credential-network combined	7	1.000	–	combined credential + egress behavior

Category	Run N	DSR	FPR	Notes
Intent-proof floor	37	0.973	–	one remaining miss in floor-policy timing
Lateral movement	1	1.000	–	nc-based lateral-probe preparation behavior
Adversarial evasion	54	0.963	–	most cases detect; two LotL/timing misses remain
Noisy benign	25	–	0.880	dominant benign false-positive source in the current live set

Categories E (Posture & Policy) and F (LAN Awareness) are validated via integration scripts with 100% pass rate (Section 7.7).

Analysis Observed false negatives (3). Most misses occur in the adversarial-evasion bucket (2 of 3), with one remaining miss in intent-proof-floor timing. That concentration is more informative than the raw count: the system remains strong on process-lineage, credential-access, combined credential-egress, lateral-movement, and core network-intent categories, while the residual miss surface is localized to living-off-the-land executions that preserve expected binaries/destinations and to cases where Layer-7 context arrives after the initial decision window.

Layer-7 timing sensitivity. The residual intent-proof-floor miss still looks like an observability-boundary issue rather than a rule-construction failure. When the 5-minute Layer-7 full-refresh cycle has not yet captured process-lineage context at snapshot time, the first decision can be too early; once enrichment lands, the same underlying logic succeeds.

Operational false positives (benign ambiguity). False positives are still localized rather than uniform: 22 of the 36 FP arise in the noisy-benign bucket (`FPR=0.880`, 22/25 runs), with the remaining 14 in the broader network-intent-correlation category. This suggests that the current burden is split between background traffic and benign-but-under-specified allowlist contexts rather than generalized instability across all categories. In deployment, that still matters:

persistent benign alerts erode operator trust quickly, so environment-specific allowlisting or baseline learning is needed before policy hardening.

High-level takeaway: the current 12-seed live evidence set reinforces the system's value as a runtime correlation layer for lineage, credential access, lateral movement, and undeclared egress, while sharpening two caveats. First, benign background traffic still needs better baseline treatment to avoid operator fatigue. Second, adversarial/LotL behavior that stays within expected tools or appears before Layer-7 enrichment remains difficult. The system is therefore best understood as a practical early-warning layer, not as an interception-grade prevention mechanism.

7.6 Ablation Studies

To interpret the overall Section 7.5 result, we run ablations that isolate which components produce most of the detection lift and where misses originate. The benchmark runner supports three ablation modes via `--ablation`:

Ablation	Disabled	Expected Impact
no-introspection (ablation flag name)	CVE vulnerability detection	Cat. D degrades to CLEAN
no-l7	Layer-7 enrichment	Cat. B, C lose detection
no-settling	Settling window	Transient misses; timing evasion

These ablations define component contributions. Running each mode writes comparative summaries to `artifacts/live-paper-summary.json` for the benchmark-summary visualization used in companion materials. In this draft, full `no-floor` / `no-l7` / `no-settling` comparative tables are not yet included in the canonical confusion-matrix snapshot.

eBPF vs netstat2 fallback. `tests/test_l7_ebpf_ablation.sh` runs the same Layer-7 checks twice: with eBPF kprobes (default on kernels ≥ 5.8), then with tracefs unmounted to force netstat2 `/proc/net/*` fallback.

Mode	Checks	Pass	Layer-7 Latency	Sessions
eBPF (kprobe)	9	9/9	15 s	50
netstat2 (fallback)	9	9/9	45 s	31

Both modes pass 9/9 and produce the same enrichment fields (`pid`, `process_name`, parent lineage including `parent_script_path`, `spawned_from_tmp`, `open_files`, cwd, memory, cpu_usage). Latency differs (15 s with eBPF vs 45 s fallback) because eBPF only accelerates socket-to-PID mapping. After PID resolution, both paths share the same `/proc` enrichment logic. Fallback can

miss very short-lived sockets between polls, but for connections that survive at least one poll cycle, detection capability is preserved.

7.7 Multi-Signal Observer Validation

After isolating component-level effects, we verify observer breadth: the system plane is not only a network monitor, but a multi-signal telemetry surface. Categories E (Posture) and F (LAN) validate that the system-plane observer extends beyond network sessions. The test suite verifies MCP tool accessibility for five signal families:

1. **Network sessions:** `get_sessions`, `get_exceptions` (Categories A-D, G-H)
2. **Anomaly/blacklist:** `get_anomalous_sessions`, `get_blacklisted_sessions`
3. **Posture:** `get_score`, `check_policy`, `advisor_get_todos` (Category E)
4. **LAN:** `get_lan_devices`, `get_lan_host_device`, `set_lan_auto_scan` (Category F)
5. **Identity:** `get_pwned_emails`, `get_breaches` (Category E)

For benchmark/tool-access validation, the suite calls `get_sessions`; the internal divergence and vulnerability loops themselves consume `get_current_sessions` within the EDAMAME host runtime.

All five families are exercised deterministically (no LLM) with 100% pass rates, confirming that the two-plane correlation loop can incorporate signals from any of these dimensions without infrastructure changes.

7.8 Empirical Validation: Four-Tier Test Architecture

Beyond the benchmark suite, we maintain a four-tier test architecture:

Tier 0: Smoke and component validation (`smoke.sh`, `test_skill_components.sh`, `test_extrapolator_validation.sh`, `test_posture.sh`). Deterministic checks for daemon health, MCP tool visibility, session APIs, behavioral model engine handoff, posture skill runtime. 79+ checks across 4 test groups, all passing.

Tier 1: Layer-7 telemetry validation (`test_l7_telemetry.sh`, `test_l7_ebpf_ablation.sh`). Deterministic checks for parent lineage, open-files detection, field correctness, and coverage (20 checks plus 2x9 eBPF/fallback checks).

Tier 2: Attack and policy/LAN detection (`test_attack_detection.sh`, `test_posture_policy.sh`, `test_lan_awareness.sh`). Engine-driven scenarios use the hybrid Rust-deterministic + LLM-adjudication pipeline for attack detection, alongside deterministic policy and LAN checks to validate posture/todos and LAN/anomaly/blacklist/HIBP APIs.

Tier 3: Two-plane E2E and vulnerability detection (`test_two_plane_e2e.sh`, `test_vulnerability_detection.sh`). Full worker + extrapolator + inject +

detector pipeline validation plus CVE-inspired vulnerability-detection checks across token exfiltration, exposed gateways, weaponized skills, sandbox escapes, credential sprawl, and memory poisoning.

Tier	Script	Checks	LLM?	Pass Rate	Notes
0	`test_poc.sh`	29	Mixed	97% (28/28 + 1 skip)	Infrastructure, telemetry, MCP tools, agent
0	`smoke.sh`	14	No	100%	Daemon health, eBPF kprobes
0	`test_skill_components.sh`	55	No	100%	MCP tools, engine handoff
0	`test_extrapolator_validation.sh`	–	Yes	100%	Behavioral model pipeline
0	`test_posture.sh`	24	Yes	100%	Skill runtime invocation
1	`test_l7_telemetry.sh`	20	No	100%*	PID, process name, cmd attribution
1	`test_l7_ebpf_ablation.sh`	2x9	No	Partial	eBPF vs fallback comparison
2	`test_attack_detection.sh`	22	Mixed	Partial**	Engine verdict pipeline under stabilization
2	`test_posture_policy.sh`	8	No	87%	7/8 pass
2	`test_lan_awareness.sh`	6	No	83%	5/6 pass
3	`test_two_plane_e2e.sh`	–	Yes	Partial**	Worker + extrapolator + detector pipeline
3	`test_vulnerability_detection.sh`	66	Yes	86%	57/66 pass

*L7 Tier 1: Basic L7 attribution (PID, process_name, full command line) passes consistently with eBPF kprobe programs active (`track_connect_v4/v6`, `trace_udp_send`, `trace_sendto`). Extended enrichment fields (parent_process_path, open_files) are populated asynchronously by the L7 enricher and depend on `/proc` scan depth; these fields are not tested in the core validation suite since they are additive context for the divergence engine rather than structural correctness requirements.

**Engine verdict: The hybrid Rust/LLM divergence engine returns structured verdicts only after a behavioral model has been ingested and the 60-second settling window has elapsed. The `test_poc.sh` suite correctly handles this as a SKIP condition when no agentic tick has occurred. Tier 0 component tests (55/55) confirm all engine MCP tools are reachable.

7.9 Discussion

Latency is telemetry settling, not model delay. The observed 40-60s TTD is dominated by EDAMAME session aggregation and Layer-7 refresh cadence. Without settling-window checks (Section 3.6), many runs would be scored on

stale data.

Process attribution is load-bearing. Attack and benign runs can generate identical traffic to the same endpoint; only process attribution separates `curl` from `python3` at the same IP. Without Layer-7 enrichment, separation collapses.

Claim discipline is mandatory. We do not claim comprehensive coverage. BadAgentUse validates five signal families and explicitly labels three designed blind spots. The suite is extensible via new JSON scenarios, without code changes.

Operational burden of benign noise. The noisy-benign slice currently shows `FPR=0.40` (2/5). This is acceptable for controlled benchmarking but high for always-on production operations; reducing benign-noise alerts is a deployment priority alongside improving attack recall.

Comparative baselines are currently absent. BadAgentUse does not yet include head-to-head comparisons against AgentSentinel, AgentArmor, LlamaFirewall, Pro2Guard, or Agent-C. This benchmark validates internal correctness of the correlation protocol and telemetry surface, not superiority. Direct comparison requires adapting either the harness for interception-style monitors or those systems for MCP telemetry (Section 9.2).

8. Reproducibility

All source code, scenario definitions, test scripts, and evaluation harnesses are published at: **https://github.com/edamametechnologies/openclaw_security**

Key artifact paths:

- `tests/benchmark/live-scenarios/`: versioned JSON scenario contracts for live runs
- `tests/benchmark/run_live_suite.py`: benchmark harness
- `tests/benchmark/summarize_results.sh`: metric computation from NDJSON traces
- `reproduce_live.sh`: single-command full reproduction
- `artifacts/`: generated results (summary JSON, NDJSON traces, paper bundle)
- `artifacts/exec-demo-pack-<timestamp>/`: latest 4-scenario executive demo artifact pack (MP4 + verification contracts)
- `docs/CLAIM_ARTIFACT_INDEX.md`: canonical claim-to-artifact lineage binding

Single-command reproduction for paper metrics:

```
./reproduce_live.sh
```

This generates:

- `artifacts/live-paper-results.ndjson`
- `artifacts/live-paper-summary.json`

- `artifacts/live-paper-manifest.json`
- `artifacts/paper-bundle-live/*` (generated whitepaper bundle with embedded metrics)

9. Limitations and Future Work

9.1 Current Limitations

1. **Living-off-the-land blind spot.** The primary residual risk is LotL behavior (Section 4.3): allowed processes exfiltrating to allowed destinations. BadAgentUse explicitly labels 3 of 50 scenarios as `DESIGNED_MISS` for this reason. Closing this gap requires behavioral baselines, anomaly fusion, or payload-aware inspection.
2. **Layer-7 refresh latency.** Full Layer-7 enrichment refresh runs every 5 minutes (sensitive-file scan every 30 s on Linux). Correlation rules that rely on `spawned_from_tmp` or `open_files` can miss short-lived processes if snapshots are stale. The `no-settling` ablation (Section 7.6) quantifies this.
3. **Reactive detection delay (operationally significant).** Example Rule A is post-observation and therefore delayed (median 45.8 s). In high-consequence environments, this window may still allow harmful actions before alerting. Example Rule B reduces this for MCP-gated actions, but direct shell/library paths can bypass pre-action checks.
4. **Interception trade-off vs AgentSentinel.** AgentSentinel reduces timing exposure for intercepted operations by pausing execution inline; our approach prioritizes portability and lower integration friction. This is a real design trade-off, not a superiority claim.
5. **Governance boundary (Cases B/C).** If the reasoning plane truthfully declares a harmful action and policy allows it, correlation alone is insufficient (Section 4.2). Authorization architecture, change control, and approvals remain mandatory.
6. **Reasoning-plane observation (partially addressed).** The extrapolator reads session transcripts via platform APIs, providing independent observation of tool calls and arguments. However, this is transcript-only – chain-of-thought reasoning, internal model state, and unlogged deliberation are not observable. A sufficiently sophisticated attacker could craft tool calls that appear benign in the transcript while achieving malicious outcomes through side effects invisible to the session record.
7. **Observer integrity boundary.** Section 4.3 assumes a trustworthy observer. Under kernel-level compromise, telemetry can be suppressed or forged.
8. **Single-platform benchmark.** BadAgentUse is currently Linux-only (aarch64 Lima VM). Multiplatform support exists in the observer but is not yet fully benchmarked in this protocol.
9. **Single-observer, non-federated focus.** The current EDAMAME runtime already supports multiple reasoning-plane contributors on

one observer-hosted engine (for example OpenClaw plus Cursor slices merged into one correlation window), but multi-host delegation still requires federated observation and cross-host correlation that are not yet implemented.

10. **No formal safety proofs.** Section 5 provides transparent rule properties, not PAC bounds or cryptographic proofs.
11. **High-impact coding-agent scenarios remain under-tested.** We have not yet fully benchmarked destructive git workflows and secret-exposure paths (e.g., accidental commit/exfiltration of repository credentials and CI tokens), where delayed detection may be especially costly.
12. **External validation gap.** Independent replication remains necessary for claims beyond controlled-environment demonstration.

Benchmark-scope note: the aggregate metrics in Section 7 quantify EDAMAME's downstream divergence/correlation performance under the explicit-slice producer mode and do not directly measure transcript-to-model accuracy of either the OpenClaw `edamame-extrapolator` skill or EDAMAME's internal raw-session ingest path. End-to-end reasoning-plane publication is validated separately by the integration and E2E paths described in Section 7.7.

9.2 Near-Term Work

- **Comparative evaluation on shared benchmarks**: run BadAgentUse scenarios against AgentSentinel, LlamaFirewall, and other monitors to produce head-to-head DSR/FPR comparisons. Requires adapting either the benchmark harness or the competing systems to a common telemetry interface.
- **Compound multi-stage attacks**: scenarios where a single compromised action chain touches all five signal dimensions (network, Layer-7, posture, LAN, identity), validating cross-dimension divergence detection in a single evaluation.
- **Adaptive evaluation**: extend BadAgentUse with scenarios that adapt their evasion strategy based on the monitor's known detection capabilities, following AgentSentinel's adaptive attack framework.
- **Reasoning-plane auditing**: add a lightweight CoT consistency check as a complementary signal. Before accepting an intent declaration, evaluate whether it is consistent with the conversation context and previous declarations.
- **Payload-aware detection**: extend Example Rule A to consider data volume and connection duration as additional signals for LotL detection.
- **MCP-native poisoning scenarios**: extend BadAgentUse with scenario families inspired by MCPTox (tool-description poisoning) and MSB (tool-response injection). The two-plane approach should detect policy-inconsistent egress or file access induced by such attacks even when the MCP call itself appears legitimate, but this requires empirical validation.

- **Observer-agnostic interface**: define an abstraction layer so that alternative telemetry providers (osquery, raw eBPF, Sysmon) can replace EDAMAME Posture, enabling independent replication and reducing the single-vendor dependency.
- **Ablation baselines**: implement a telemetry-only baseline (B0: blacklist + anomaly score only, no intent gating) and an allowlist-only baseline (B1: no critical signals) to quantify the contribution of intent correlation beyond existing observer signals.
- Measure rollback reliability under reversible action protocols.
- Multi-host replication with federated observers.
- Cross-platform benchmark runs on macOS and Windows (EDAMAME Posture supports both platforms; the benchmark harness requires adaptation).

10. Ethics and Safe Use

This system provides defensive runtime monitoring; it does not grant authorization decisions or replace IAM/policy controls. It should be deployed alongside baseline hardening controls (S1 and S2), manual confirmation paths for high-impact actions, and claim discipline tied to measured evidence and protocol scope.

10.1 Telemetry Privacy and Data Sovereignty

EDAMAME Posture follows a **privacy-first, user-up** architecture: all telemetry is collected, processed, and stored locally on the endpoint. Session data, Layer-7 enrichment, and posture snapshots are retained for a rolling 8-hour window and then discarded. No telemetry – including process metadata, open file paths, command lines, or network sessions – is ever transmitted to the EDAMAME backend or any external service. The EDAMAME platform (https://edamame.tech) is designed so that endpoint security operates in a user/agent-up model where the user or agent owns and controls their own security posture; no administrator can look down into endpoints or collect personally identifiable information (PII). This stands in contrast to traditional enterprise EDR/MDM architectures where a central admin has full visibility into endpoint activity.

In multi-user or shared-host deployments, Layer-7 enrichment may include usernames and command-line arguments, creating a local privacy surface. The MCP endpoint defaults to localhost binding with PSK authentication (Section 10.2), ensuring telemetry is not exposed beyond the local host.

10.2 MCP Endpoint Exposure

The Streamable HTTP MCP endpoint defaults to localhost binding (127.0.0.1:3000). Misconfiguration – binding to 0.0.0.0 or routing through a reverse proxy without authentication – would expose sensitive runtime

telemetry to the network. EDAMAME Posture supports pre-shared key (PSK) authentication; we recommend enabling it whenever the endpoint is reachable beyond loopback. The observer process itself is part of the attack surface (Section 4.3): an adversary who compromises the MCP endpoint can suppress alerts or inject false telemetry. Deployment hardening should follow the MCP specification's transport security guidelines.

10.3 Ethical Considerations (Menlo Report Alignment)

We map our research to the Menlo Report's four ICT research ethics principles:

- **Respect for Persons.** No human subjects or real-user data are involved. All evaluation runs in an isolated Lima VM with synthetic injections. No PII is collected, stored, or transmitted. The 8-hour local retention window (Section 10.1) ensures no persistent personal data accumulates.
- **Beneficence.** The system is purely defensive. We document blind spots (LotL evasion, Section 4.3) and design limitations to prevent false confidence.
- **Justice.** Benchmark scenarios, decision rules, and evaluation scripts are open-source for independent replication and critique.
- **Respect for Law and Public Interest.** Telemetry collection occurs only on the researcher's own VM. No third-party systems are probed or scanned. Benchmark injections target localhost or controlled endpoints only.

10.4 AI Tools Disclosure

Generative AI tools (Claude, Cursor AI) assisted with code scaffolding and paper editing. All outputs were reviewed and validated by the human authors. No AI system is listed as an author. Scenario definitions, decision rules, and architecture diagrams are human-authored. LLM-driven test tiers (Categories A–C) use `openai/gpt-5.1` (with compatible `azure-openai-responses/*` alternate provisioning); deterministic tiers do not use LLM inference. Verdicts are deterministic given fixed telemetry (Example Rule A); only the natural-language explanation varies across runs.

11. Conclusion

We present a practical two-plane runtime security model: correlate agent intent (reasoning plane) with host behavior (system plane), and flag divergence when behavior is unexplained by intent or triggers intent-independent critical signals (CVE vulnerability detection, blacklisted traffic). We instantiate the model in OpenClaw with EDAMAME Posture and evaluate it using trace-backed live artifacts.

The evidence supports the architectural claim: process-attributed telemetry plus intent correlation yields operationally useful detection. The boundary condi-

tions are explicit: living-off-the-land behavior and telemetry timing windows remain primary blind spots, and comparative baseline studies are future work.

Appendix A. Multi-Signal Scenario Sketches

These sketches illustrate how the same two-plane logic extends beyond pure egress checks when richer telemetry is available.

A1. Lateral movement (LAN + traffic). A compromised skill performs local recon and attempts service probing. LAN discovery APIs detect new hosts/open ports while network sessions attribute traffic to the originating process tree. If reasoning-plane intent remains "document summarization," the mismatch yields divergence.

A2. Credential theft (identity + traffic). A supply-chain compromise accesses credential stores and initiates outbound transfer. Identity/breach signals increase risk context while Layer-7 open-file/process metadata and network telemetry provide direct evidence for correlation.

A3. Compound attack (network + Layer-7 + identity + posture). A staged workflow scans, reads sensitive material, and exfiltrates. No single signal family is sufficient, but cross-signal correlation yields stage-specific verdicts and supports clearer operator triage.

Appendix B. Portability Notes and Future Work

Section 3.4 and Section 3.5 present the model-level portability argument in the main narrative. This appendix keeps that discussion intentionally brief: it documents the OpenClaw-specific correspondence and then scopes portability work that remains future research.

B.1 How the Skill Architecture Differs from Orchestration Frameworks

Most agentic stacks are built on orchestration frameworks: LangGraph (LangChain, 2024), CrewAI (Moura, 2024), and AutoGen (Microsoft, 2023). These frameworks share several characteristics:

- **Graph-based execution.** Workflows compile into DAG/state-machine steps.
- **Cloud-first telemetry.** Traces often flow through external observability services (e.g., LangSmith, AgentOps, Arize Phoenix).
- **Framework coupling.** Guardrails are usually callback/runtime specific.

The OpenClaw skill model is architecturally distinct on every axis:

Property	Orchestration frameworks	OpenClaw skill model
Execution model	Compiled graph / state machine	Conversational Markdown instructions
Runtime state	Framework-managed checkpoints	Agent-controlled memory files
Security boundary	Framework callbacks / middleware	Declarative manifest (`clawhub.json`)
Telemetry path	Cloud API (LangSmith, AgentOps)	Local-only MCP (localhost, PSK)
Agent coupling	Framework-specific (LangGraph nodes, CrewAI tasks)	Protocol-level (MCP tools)
Dependency weight	Heavy (Python ecosystem, SDK, cloud auth)	Zero-framework (Markdown + MCP HTTP)

In OpenClaw, the skill is Markdown (`SKILL.md`) interpreted at runtime, without compiled workflow graphs or framework SDK dependencies. This keeps the security contract independent of framework internals.

B.2 Skill-to-Architecture Correspondence

The OpenClaw-facing reasoning skill and the EDAMAME-internal loops (divergence plus posture/remediation) interact with three EDAMAME layers, but through different interfaces depending on whether the component runs inside or outside the observer process:

Telemetry layer. In the reference benchmark deployment, EDAMAME Posture (via flodbadd) provides continual, process-attributed sessions with Layer-7 enrichment. On workstation-hosted Cursor paths, the same internal runtime is hosted by the EDAMAME Security app. The internal divergence and vulnerability loops access this data directly via Rust method calls (`get_current_sessions`, `get_anomalous_sessions`, `get_blacklisted_sessions`, `get_lanscan`) within the host runtime; no MCP round-trip is involved. This is a deliberate security property: the telemetry path from capture to verdict is entirely internal, unreachable by external processes (Section 4.4). The extrapolator skill does not read EDAMAME telemetry at all; it reads only OpenClaw session transcripts via platform APIs (`sessions_list`, `sessions_history`) and publishes its behavioral model to EDAMAME through the `upsert_behavioral_model` MCP tool.

Scoring layer. EDAMAME Foundation computes composite scores and compliance mapping (CIS, ISO 27001, SOC 2). The internal engine reads score directly (`get_score`) for posture-delta detection. The optional posture facade can read the same data via MCP (`get_score`, `advisor_get_todos`) while reusing the same native remediation pipeline.

Action layer. EDAMAME's agentic pipeline (`agentic_execute_action` / `advisor_undo_action`) provides execute-and-rollback with confirmation gates. The internal posture loop owns scheduled remediation, while the optional posture facade can consume this layer via MCP for explicit invocations. Example Rule B intercepts here to enforce intent consistency before actions execute.

The key insight is architectural alignment with security isolation: the OpenClaw-facing reasoning skill interacts with EDAMAME through authenticated MCP endpoints, while divergence and posture/remediation loops run inside the observer and access telemetry directly. This separation means that compromising the reasoning plane cannot tamper with the telemetry-to-verdict path, while MCP remains the stable, portable interface for external consumers.

B.3 Future Work: Extending Beyond OpenClaw-Class Flows

This paper evaluates a specific class of agent behavior: cooperating single-node conversational workflows sharing state through memory files. We do **not** claim implemented portability to graph-orchestrated, multi-node, or multi-agent frameworks in this draft.

Future work priority areas are:

1. **Node-scoped intent declarations.** Move from session-scoped intent to per-node intent so allowlists can change safely across workflow steps.
2. **Workflow-aware correlation.** Attribute observations to the correct node across transitions and retries, rather than only to the currently active step.
3. **Delegation trust chains.** Verify that delegated intents are constrained by parent intent and do not expand permission budgets across agent boundaries.
4. **Monitorable workflow design.** Enforce explicit node intent, minimal per-node tool scope, observable transitions, bounded delegation depth, and immutable compiled graphs.
5. **Observer-agnostic replication.** Standardize interfaces so alternative observers can provide equivalent system-plane telemetry for independent studies.

B.4 Scope Boundary for This Draft

Accordingly, this manuscript does not claim:

- validated LangGraph/CrewAI/AutoGen portability results,
- formal workflow-level safety proofs for multi-node execution, or
- cloud-governance analysis for deployments where telemetry leaves endpoints.

These remain future-work items after a workflow-aware prototype exists and can be evaluated with trace-backed scenarios.

12. Author Disclosure and Competing Interests

All two authors have financial relationships with EDAMAME Technologies (founder, board member respectively). The bounding argument is that the evaluation target is the correlation method over published artifacts, not proprietary internals, and that independent replication is practically enabled by open code, trace-level evidence, and observer replaceability.

Frank Lyonnet is the founder of EDAMAME Technologies, the company that develops EDAMAME Posture. This paper evaluates EDAMAME Posture as the system-plane observer – a core architectural dependency. Antoine Clerget is a board member of EDAMAME Technologies.

To bound this conflict: (a) the observer is treated as a black-box telemetry provider, not the evaluation target; (b) decision rules, scenarios, harness, and scripts are open-source; (c) every verdict is trace-backed with on-disk artifacts; and (d) the core observer repositories are open-source: **flodbadd** (session capture + Layer-7 attribution), **edamame_foundation** (scoring/threat infrastructure), and **threatmodels** (endpoint threat-model definitions). All are published under EDAMAME Technologies and can be audited or replaced by alternative observers. We actively invite independent replication.

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